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Clinical paper

Socioeconomically equitable public defibrillator placement using mathematical optimization



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Abstract

Background: Mathematical optimization can be used to place automated external defibrillators (AEDs) in locations that maximize coverage of out-of-hospital cardiac arrests (OHCAs). We sought to determine whether optimization can improve alignment between AED locations and OHCA counts across levels of socioeconomic deprivation.

Methods: All suspected OHCAs and registered AEDs in Scotland between Jan. 2011 and Sept. 2017 were included and mapped to a corresponding socioeconomic deprivation level using the Scottish Index of Multiple Deprivation (SIMD). We used mathematical optimization to determine optimal locations for placing 10%, 25%, 50%, and 100% additional AEDs, as well as locations for relocating existing AEDs. For each AED placement policy, we examined the impact on AED distribution and OHCA "coverage" (suspected OHCA occurring within 100 m of AED) with respect to SIMD quintiles.

Results: We identified 49,432 suspected OHCAs and 1532 AEDs. The distribution of existing AED locations across SIMD quintiles significantly differed from the distribution of suspected OHCAs ($P < 0.001$). Optimization-guided AED placement increased coverage of suspected OHCAs compared to existing AED locations (all $P < 0.001$). Optimization resulted in more AED placements and increased OHCA coverage in areas of greater socioeconomic deprivation, such that resulting distributions across SIMD quintiles matched the shape of the OHCA count distribution. Optimally relocating existing AEDs achieved similar OHCA coverage levels to that of doubling the number of total AEDs.

Conclusions: Mathematical optimization results in AED locations and suspected OHCA coverage that more closely resembles the suspected OHCA distribution and results in more equitable coverage across levels of socioeconomic deprivation.

Keywords: Out-of-hospital cardiac arrest Automated external defibrillators Socioeconomic factors Scottish Index of Multiple Deprivation

Introduction

Out-of-hospital cardiac arrest (OHCA) is a major health issue that affects 6–7 million adults around the world annually.^{1–3} The odds of OHCA survival to hospital discharge decrease by 7–13% per minute without treatment.⁴ Despite improvements over the past 15 years, the likelihood of survival remains low overall.^{5,6} Early initiation of cardiopulmonary resuscitation and defibrillation are integral components of the 'chain of survival' for OHCA response.^{7,8}

Public access defibrillation (PAD) programs are effective at increasing rates of bystander-initiated defibrillation and survival.^{9–11} However, defibrillator accessibility remains a significant concern since OHCAs typically occur far from automated external defibrillators (AEDs) and increased travel distance to an accessible AED is independently associated with poorer outcomes for OHCA victims.^{12–14} These findings highlight the need for AED placement methodologies that improve AED proximity to prospective OHCA locations.

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Mathematical optimization is an effective approach to maximising the spatiotemporal accessibility of AEDs,^{15–19} and has been shown to outperform both actual and guideline-based AED placement.^{18,19} However, the effect of optimization on the distribution of AED placement across socioeconomic deprivation levels is unknown. Related work to date has examined the socioeconomic distribution of existing AED placements, rather than optimal AED locations.^{20–22} Understanding the distribution of AEDs across areas of varying socioeconomic deprivation is important because socioeconomic deprivation has been associated with greater incidence of OHCA,^{20,22–24} lower likelihood of receiving bystander-initiated resuscitation,²⁵ lower AED availability,^{21,22} and poorer outcomes.^{24,26,27}

In this study, we examine the distribution of mathematically optimized AED locations across socioeconomic groups using data from Scotland. We are particularly interested in whether optimization can improve the alignment between AEDs and OHCA across communities with differing levels of socioeconomic deprivation.

Methods

This study was approved by the Research Ethics Board at Queen's University at Kingston (protocol number: 6021230). This study was conducted as part of the Scottish Government's OHCA Strategy, a five-year plan published in 2015 aimed to investigate, develop, and implement data-driven policies to improve outcomes for OHCA victims.²⁸

Study setting

Scotland is a constituent country within the United Kingdom with an area of 77,937 km² and a population of 5.35 million. The country is divided into 31 Health and Social Care Partnerships, which are organisations responsible for the delivery and coordination of health and social services to residents, and are jointly run by National Health Service Scotland and local councils.²⁹

Scotland is also divided into 6976 data zones on the basis of national census findings. Each zone contains between 500 and 1000 residents and was assigned a rank according to the 2016 Scottish Index of Multiple Deprivation (SIMD). SIMD is a tool developed by the Scottish Government for identifying areas of poverty and inequality across Scotland aggregated from 38 indicators across the domains of household income, employment, education, health, access to services, crime, and housing.³⁰ Within each domain, data zones are first scored and ranked based on indicator values; the domain rank is then exponentially transformed. Each data zone is then given an aggregate SIMD score (weighting: household income 28%, employment 28%, education 14%, health 14%, access to services 9%, crime 5%, housing 4%), and finally given an overall SIMD rank. Based on their SIMD rank, data zones are divided into quintiles, with 1 representing the most deprived quintile and 5 representing the least deprived. Because each data zone is scored independently, less deprived data zones may be directly adjacent to more deprived data zones.

Study population

All incidents involving Scottish Ambulance Service dispatch for suspected OHCA between January 1, 2011 and September 30, 2017 occurring in residential or public settings were considered for inclusion in the study; this includes both treated and untreated OHCA. Incidents missing location information were excluded. All existing

AEDs in Scotland registered with the Scottish Ambulance Service as of May 2018 were included in the analysis. AED registration is voluntary in Scotland, and to our knowledge, coordinated AED placement programmes on a large scale do not currently exist.

Analysis

We computed the frequency distributions across SIMD quintiles of historical suspected OHCA and existing AEDs. Within each frequency distribution, we determined whether the proportions differed across the SIMD quintiles using the Kruskal-Wallis test. We also determined whether the frequency distribution of suspected OHCA differed from the frequency distribution of AEDs using the chi-squared goodness-of-fit test.

To determine optimal AED locations for each Health and Social Care Partnership region, we applied a previously validated mathematical optimization model.¹⁵ Candidate AED locations were generated throughout each region using a 50 m × 50 m grid. The model sought to select optimal candidate locations for new AEDs to maximize overall coverage of suspected OHCA with the Health and Social Care Partnership region across both existing and new AED locations. We considered an OHCA covered by an AED if there was an AED within 100 m walking distance following the road and pedestrian network, which was calculated using the QGIS 3.10 geographical analysis software (Open Source Geospatial Foundation). The model was constrained to hold fixed the locations of the “*N*” existing AEDs in each Health and Social Care Partnership region and determine the optimal placements of “*R*” new AED locations. We chose values of *R* equal to 10%, 20%, 50% or 100% of *N*, rounded down to the nearest integer. We determined the values of *N* and *R* for each of the 31 Health and Social Care Partnership regions based on the number of existing AEDs in that region.

To compare coverage performance between optimized and existing AED placements, we conducted a secondary analysis by modifying the optimization model to determine *N* AED locations, but allowing them to be placed without regard for the existing AED locations. Semantically, this is allowing for the relocation of existing quantities of AEDs in an optimal manner, without the addition of any new AEDs. All models were solved using Gurobi 9.0 optimization software (Gurobi Optimization LLC, Houston, TX, United States) within the Python 3.7 software language (Python Software Foundation).

We evaluated the performance of the model on out-of-sample suspected OHCA using five-fold cross validation. First, we divided the suspected OHCA dataset for each Health and Social Care Partnership region into five disjoint subsets. We applied the optimization models to each region using four of the five subsets to determine optimal AED locations; each region was considered separately in order to keep the optimization models computationally tractable. These locations were then used to measure coverage of suspected OHCA in the fifth (out-of-sample) subset. We repeated this process four more times, once for each additional out-of-sample subset. We then computed the frequency distributions of optimized AEDs and the resulting coverage of out-of-sample suspected OHCA across the SIMD quintiles, and aggregated these results on a national scale.

Comparing the optimized and existing AEDs, we determined whether the proportion of covered suspected OHCA differed using the Wilcoxon signed-rank test, and whether their distributions across the SIMD quintiles differed using the chi-squared goodness-of-fit test. For all statistical tests, a value of *P* < 0.05 was considered sig-

nificant. Statistical analyses were conducted using the R 3.6.1 software language (R Foundation for Statistical Computing).

Results

The Scottish Ambulance Service responded to a total of 49,752 suspected OHCA during the study period. Of those, 260 responses were duplicate dispatches for the same event, 23 incidents did not have an available incident location and 37 incidents occurred outside of Scotland; these 320 incidents were excluded from the analysis, resulting in a total of 49,432 suspected OHCA for our analysis. We identified 1532 existing AED locations in Scotland. Table 1 shows demographic characteristics as well as the numbers of suspected OHCA and existing AEDs for each SIMD quintile.

Fig. 1 illustrates the geographical distribution of data zones classified by SIMD quintile and compares it with the distribution of suspected OHCA per square kilometre, using the city of Edinburgh as an example. A close correspondence can be seen between the areas with high socioeconomic deprivation and the areas with high numbers of suspected OHCA per square kilometre during the study period. The correspondence is similar for other cities in Scotland.

The nationwide distribution of suspected OHCA and existing AED locations across the SIMD quintiles is shown in Fig. 2. The proportions of suspected OHCA significantly differed ($P < 0.001$) across the SIMD quintiles. The proportion of suspected OHCA was highest in the SIMD 1 quintile, which represents the most socioeconomically deprived 20% of data zones, at 30.7% of all suspected OHCA during the study period, and decreased as the level of socioeconomic deprivation decreased, reaching a low of 11.2% at the SIMD 5 quintile, which represents the least socioeconomically deprived 20% of data zones.

The proportions of existing AED locations significantly differed ($P < 0.001$) across the quintiles. In contrast to proportions of suspected OHCA, the proportion of existing AED locations was highest in the SIMD 3 quintile at 30.8% of all AEDs, and decreased towards both ends of the distribution, with 10.0% of all AEDs in the SIMD 1 quintile and 15.2% in the SIMD 5 quintile. Overall, the distributions of suspected OHCA and existing AED locations significantly differed from each other ($P < 0.001$), indicating a mismatch between proportions of AED location and suspected OHCA across SIMD quintiles.

Fig. 3 compares the nationwide distribution of existing and optimized AED locations across SIMD quintiles. The distributions of AED locations across SIMD quintiles under all optimization-guided

policies significantly differed from the distribution of existing AED locations (all $P < 0.001$). Placing additional AEDs in optimized locations while holding the existing AEDs fixed did not substantially change the overall shape of the AED location distribution; however, as the number of new AEDs placed increased, AEDs were placed more frequently in the SIMD quintiles representing greater socioeconomic deprivation (i.e., SIMD quintiles 1 and 2) than in the other SIMD quintiles. When existing AEDs are optimally relocated (i.e., no additional AEDs are placed), the number of AEDs was maximized in SIMD quintile 1 and decreased towards SIMD quintile 5, matching the shape of the suspected OHCA proportion distribution from Fig. 2. Within each distribution, the number of AED locations significantly differed across the SIMD quintiles, regardless of the number of AED locations added (all $P < 0.001$).

Fig. 4 displays the nationwide distribution of out-of-sample suspected OHCA coverage across the SIMD quintiles under existing and optimized AED locations. Total coverage was greater under any optimization-guided AED placement policy (all $P < 0.001$). The coverage level distribution from the existing AEDs mirrored the distribution of existing AED locations from Fig. 2, with coverage being highest in SIMD quintile 3 at 1.2% of suspected OHCA within the quintile covered, and decreasing towards both ends of the distribution, with 0.4% of suspected OHCA within SIMD quintile 1 and 0.6% of suspected OHCA within SIMD quintile 5 covered. As the number of optimized AED locations increased, the increase in suspected OHCA coverage was higher in the SIMD quintiles representing more socioeconomically deprived areas (i.e., SIMD quintiles 1 and 2). All optimized-guided AED placement policies resulted in significantly greater OHCA coverage across SIMD quintiles than the existing AEDs (all $P < 0.001$). Among the policies that optimized additional AED locations, the policy that doubled the number of AEDs (i.e., $R = 100\%$) resulted in a coverage distribution that matched the overall shape of the suspected OHCA distribution (i.e., coverage was highest in SIMD quintile 1 and decreased towards SIMD quintile 5). When optimally relocating AEDs, the distribution of suspected OHCA coverage across SIMD quintiles had the same shape as the suspected OHCA distribution.

Discussion

To our knowledge, this is the first study to analyse how optimization-driven AED placement strategies can affect the alignment between AEDs and suspected OHCA across socioeconomic groups. The

Table 1 – Characteristics of Scottish Index of Multiple Deprivation (SIMD) quintiles. The SIMD 1 quintile represents the most socioeconomically deprived 20% of data zones, while the SIMD 5 quintile represents the least socioeconomically deprived 20% of data zones. Population is rounded to the nearest 100 persons.

SIMD Quintile	1	2	3	4	5	Total
Number of data zones	1395	1395	1395	1395	1396	6976
Population	1,049,300	1,046,200	1,066,900	1,080,200	1,105,000	5,347,600
Area (km ²)	482	10,124	38,695	25,661	2975	77,397
Population density (persons per km ²)	2177.0	103.3	27.6	42.1	371.4	69.1
Number of suspected OHCA during study period (crude)	15,279	11,771	9607	7454	5581	49,692
Number of suspected OHCA per 100,000 person-years (crude)	215.7	166.7	133.4	102.2	74.8	137.7
Number of existing AEDs	153	250	472	424	233	1532
Number of existing AEDs per 100,000 persons	14.5	23.8	44.5	40.1	21.7	28.9



Fig. 1 – A comparison between SIMD 2016 quintiles (above) and the number of suspected OHCA per square kilometre in each data zone during the study period (below) within the city of Edinburgh.

two main findings are: (1) mathematical optimization of new AED locations based on historical suspected OHCA can improve alignment between need and access across quintiles of socioeconomic deprivation, especially in more socioeconomically deprived regions; (2) optimally relocating existing AEDs provided similar benefit to adding an equal number of new AEDs.

While optimization improved suspected OHCA coverage across all socioeconomic groups, more benefit accrued to groups with greater socioeconomic deprivation. In particular, optimization placed more AEDs (Fig. 3) and increased suspected OHCA coverage (Fig. 4) the most in regions with higher socioeconomic deprivation. By using optimization to place AEDs, the resulting suspected OHCA

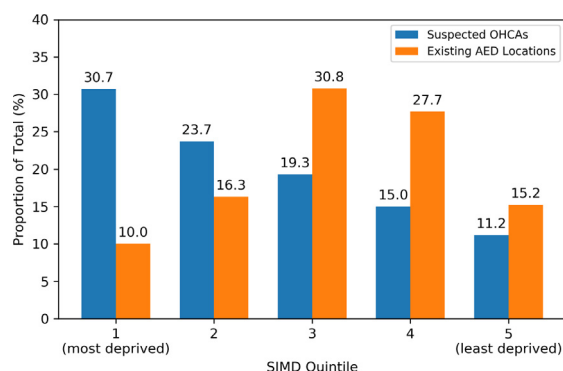


Fig. 2 – The proportions of suspected OHCA during the study period and AED locations by SIMD quintile.

coverage distribution across socioeconomic groups better resembled the shape of the suspected OHCA distribution as the number of AEDs placed increased. Although the distribution of AED locations across SIMD quintiles did not closely resemble the suspected OHCA distribution for the quantities of R considered in this study, we believe the distributions would achieve greater resemblance with larger values of R due to the greater rate of increase in AED locations in more socioeconomically deprived areas.

Previous literature suggested that mathematical optimization may result in a trade-off between efficiency and equity: spreading out resources often results in less efficient utilisation.³¹ In this study, however, we find that there is alignment between efficiency and equity: the regions with the highest level of socioeconomic deprivation are among those with the highest population density and highest rates of OHCA (Table 1). Thus, even though the optimization model did not explicitly optimize for socioeconomic deprivation, the fact that there is a strong relationship between suspected OHCA counts and socioeconomic deprivation resulted in improved equity in the AED distribution. This highlights the potential benefit of greater investment of health interventions to improve OHCA response in these areas. Beyond the placement of AEDs, this may include targeted education and social marketing campaigns in these communities.

There was a mismatch between the distribution of existing AED locations in Scotland and suspected OHCA across socioeconomic groups (Fig. 2). The first SIMD quintile had the fewest existing AEDs but the most suspected OHCA. In other words, residents of the

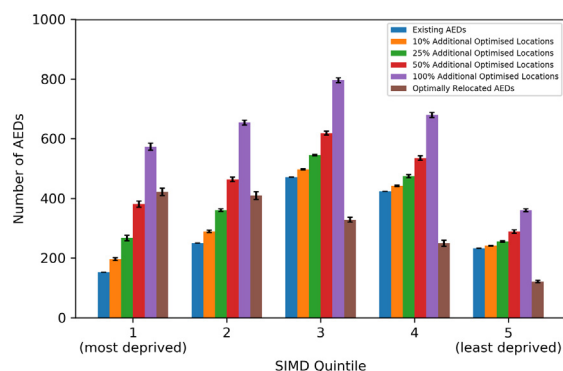


Fig. 3 – Distributions of AED locations within each SIMD quintile for each AED placement policy.

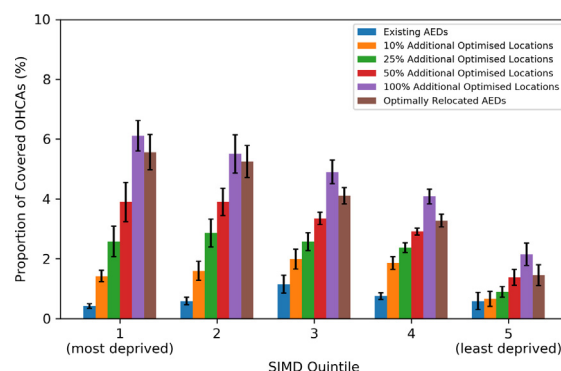


Fig. 4 – Proportions of covered suspected OHCA within each SIMD quintile for each AED placement policy.

most socioeconomically deprived areas in Scotland were not only at higher risk of OHCA but were also less likely to have an AED nearby. This finding is consistent with previous studies in other countries that find that socioeconomically deprived neighbourhoods tend to have higher counts of suspected OHCA and lower AED accessibility.^{20–23} SIMD quintiles 3 and 4 had the most AEDs, but fewer than average suspected OHCA. Table 1 shows that those quintiles cover the largest land area. As such, it appears that existing AED placements across Scotland were influenced more by geography, rather than the distribution of suspected OHCA. However, as the population within each SIMD quintile may differ in composition and risk of OHCA, the true need for AEDs may vary across quintiles.

Coverage of suspected OHCA from existing AEDs was low across all of Scotland (Fig. 4). However, it should be noted that this study considered both urban and rural locations across the entirety of Scotland; this is in contrast to prior literature on optimal AED placement models, which were limited in scope to urban settings only.^{15–19} Existing literature has also found poor proximity between suspected OHCA and existing AEDs, especially in areas with lower socioeconomic status.¹² Thus, the results in this paper reinforce the general finding that existing placement AED methods, which tend to be arbitrary and uncoordinated, do not place AEDs in the locations that maximize suspected OHCA coverage. A limitation of our data was that private and public location OHCA could not be differentiated, which may explain the low coverage levels in our study.

Our results indicate that if the existing AEDs in each Health and Social Care Partnership region could be relocated within their regions, suspected OHCA coverage levels could reach 80–90% of the coverage associated with doubling the current number of AEDs. These results suggest that the existing AED locations are highly sub-optimal and unrepresentative of historical OHCA locations, a finding consistent with previous literature.^{12,15,18} Although relocating existing AEDs is likely to be much more difficult to implement than placing additional AEDs, it would be more cost-effective. Overall, our results also suggest that centrally coordinated AED placement can lead to much greater coverage of OHCA,^{15–19} echoing the importance of data-driven PAD programs.

The study has several limitations. First, we assumed that all candidate AED locations within each Health and Social Care Partnership region are viable for AED placement. In reality, not all selected locations may be suitable due to environmental or security constraints, for example; decision makers would need to evaluate real-life feasi-

bility of optimized locations before proceeding with actual placement. However, rather than prescribing practical locations for AEDs, the goal of our model is to provide a rigorous framework to quantify the benefit of improved AED placement. Second, we focused on spatial coverage, which assumes that AEDs are available at all hours. This was done to ensure a fair comparison between existing AEDs and optimized AEDs, and because an ideal PAD program has 24/7-accessible AEDs.^{14,16,18} However, not every location can accommodate 24/7-accessible AEDs and costs associated with making AEDs available around the clock may be prohibitive in certain locations. Third, spatial coverage is not a clinical outcome, so the increased coverage due to optimization does not imply increased bystander AED use or OHCA survival. However, recent studies have shown that improved proximity of AEDs to OHCA may lead to higher rates of bystander defibrillation and survival.^{13,18,19} Finally, AED registration is voluntary in Scotland; the presence of unregistered AEDs may result in OHCA coverage levels that are greater than stated in this study. However, as the Scottish Ambulance Service can only direct responders to registered AEDs, the usefulness of unregistered AEDs would be limited to the responders' knowledge of the surrounding area, thus limiting their benefit to OHCA. Initiatives such as the British Heart Foundation's "The Circuit" may help mitigate this issue by providing a centralized AED registry accessible to ambulance services nationwide.³²

Conclusion

Mathematical optimization can identify locations for AED placement that both increase spatial coverage of suspected OHCA and better match the distribution of OHCA across levels of socioeconomic deprivation. Governments and health service organisations should centrally coordinate AED placement in order to achieve a socioeconomically equitable distribution of AEDs.

CRediT authorship contribution statement

K.H. Benjamin Leung: Methodology, Software, Data curation, Formal analysis, Investigation, Writing - original draft. **Steven C. Brooks:** Conceptualization, Writing - review & editing, Project administration. **Gareth R. Clegg:** Conceptualization, Writing - review & editing, Funding acquisition. **Timothy C.Y. Chan:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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